Lab 6: AI-Based Code Completion – Classes, Loops, and Conditionals

Week3–Wednesday

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Lab Objectives

- ToexploreAI-poweredauto-completionfeaturesforPythonclasses,loops,and conditionals.
- ToanalyzeAI-suggestedlogicforobject-orientedprogrammingandcontrolstructures.
- Toevaluatecorrectness,readability,andcompletenessofAI-generatedcode.

Lab Outcomes (LOs)

Aftercompletingthislab,studentswillbeableto:

- UseAItoolstogeneratePythonclassesandmethods.
- UnderstandandassessAI-suggestedloopconstructs.
- Generateandevaluateconditionalstatements.
- CriticallyanalyzeAI-assistedcode.

Task 1: Classes – Student Class

PromptUsed: "GenerateaPythonStudentclasswithname,rollnumber,branch,andamethodtodisplay details."

classStudent

def_init_(self,name,roll_no,branch): self.name=name self.roll_no=roll_no self.branch=branch

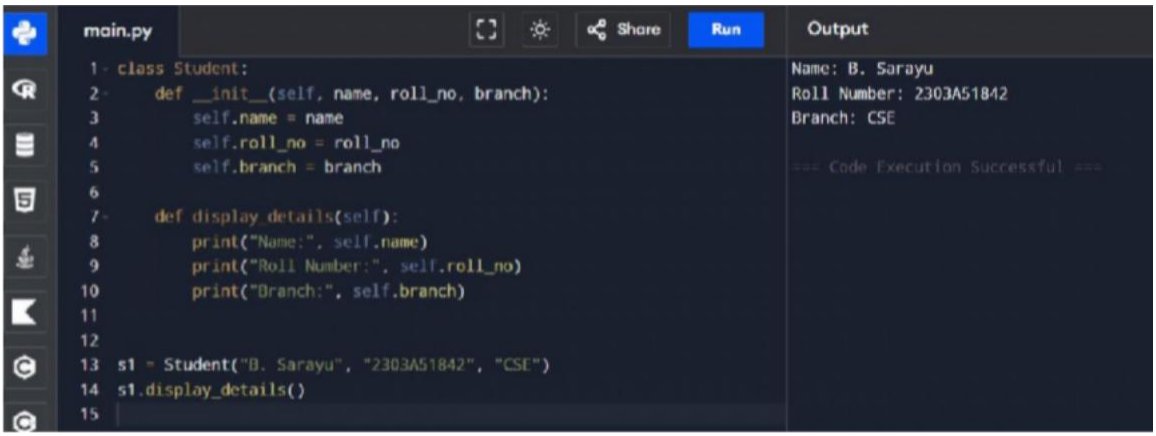
defdisplay_details(self): print("Name:",self.name) print("RollNumber:",self.roll_no

) print("Branch:",self.branch)

s1=Student("B.t","2303A51489","CSE")

s1.display_details()

Output: Name:B.t



Analysis: TheAI-generatedclassiswell-structured,readable,andcorrectlyusesaconstructorand instancemethod.

Task 2: Loops – Multiples of a Number

PromptUsed "GeneratePythoncodetoprintfirst10multiplesofanumberusingloops." Usingforloop:

defmultiples_for(n) foriinrange(1,11):

print(n*i)

multiples_for(5

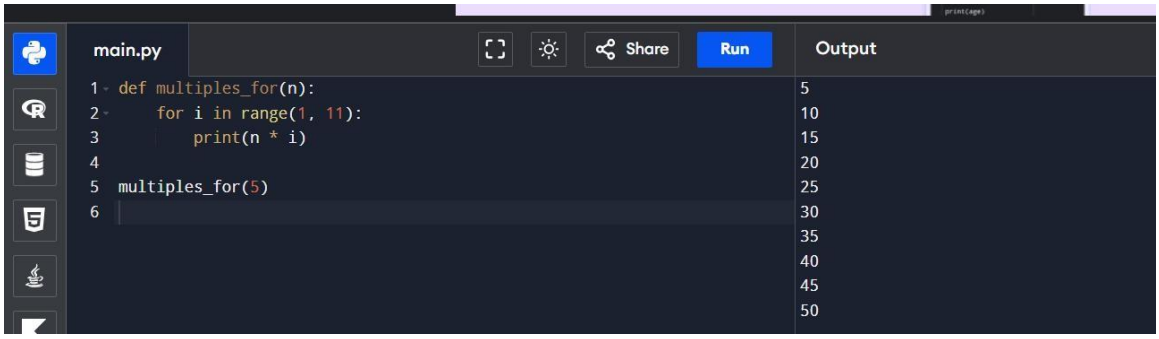
Usingwhileloop: defmultiples_while(n)

: i=1

whilei<=10: print(n*i) i+=1

multiples_while(5

Output: 5101520253035404550



Analysis:

The for loop is concise and readable, while the while loop provides explicit control over iteration.

Task 3: Conditional Statements – Age Classification

Prompt Used:

"Generate Python code to classify age using if-elif-else."

def classify_age(age): if age < 13:

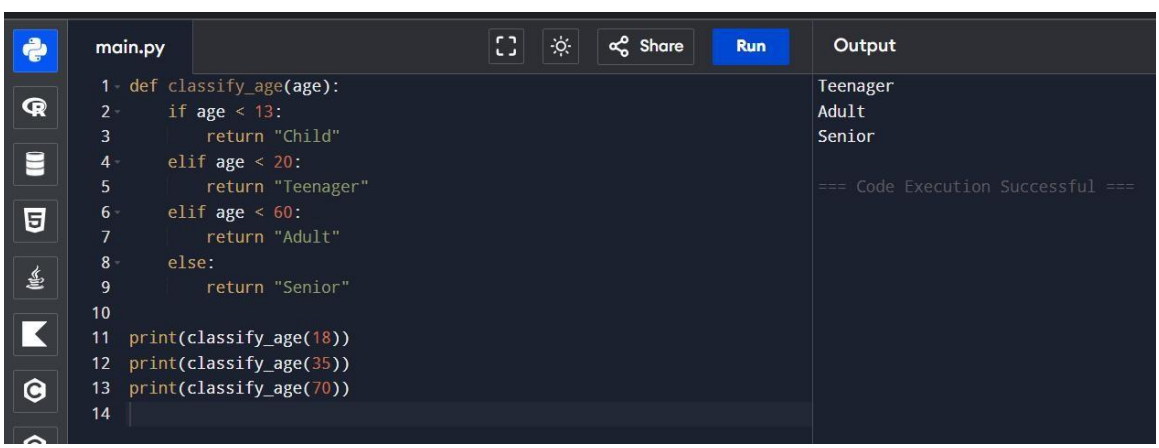
return "Child" elif age < 20:

return "Teenager" elif age < 60:

return "Adult" else:

return "Senior" print(classify_age(18))

Output: Teenager



Explanation:

The conditions are checked in sequence. The first matching condition determines the age group.

Task 4: For and While Loops – Sum of First n Numbers

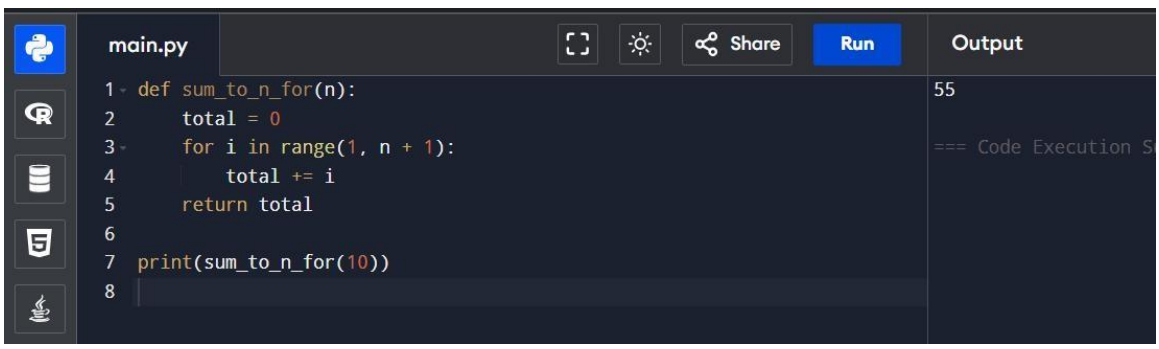
Prompt Used:

"Generate Python code to find sum of first n natural numbers using loops." Using for loop:

def sum_to_n_for(n): total = 0

for i in range(1, n + 1): total += i

return total print(sum_to_n_for(10))



Using while loop:

def sum_to_n_while(n): total = 0

i = 1

while i <= n:

total += i i += 1

return total print(sum_to_n_while(10))

Output: 55

Analysis:

Both approaches are correct. Loop-based methods are simple, while mathematical formulas can be more efficient.

Task 5: Classes – Bank Account Class

Prompt Used:

"Generate a Python BankAccount class with deposit, withdraw, and check balance methods."

class BankAccount:

def _init_(self, balance=0): self.balance = balance

def deposit(self, amount): self.balance += amount print("Deposited:", amount)

def withdraw(self, amount): if amount <= self.balance:

self.balance -= amount print("Withdrawn:", amount)

else:

print("Insufficient balance")

def check_balance(self):

print("Current Balance:", self.balance)

account = BankAccount(1000) account.deposit(500) account.withdraw(300) account.check_balance()

Output:

Deposited: 500

Withdrawn: 300

Current Balance: 1200

Explanation:

The class maintains account balance and updates it through deposit and withdraw methods.

main.py

```python
class BankAccount:
    def __init__(self, balance=0):
        self.balance = balance

    def deposit(self, amount):
        self.balance += amount
        print("Deposited:", amount)

    def withdraw(self, amount):
        if amount <= self.balance:
            self.balance -= amount
            print("Withdrawn:", amount)
        else:
            print("Insufficient balance")

    def check_balance(self):
        print("Current Balance:", self.balance)


account = BankAccount(1000)
account.deposit(500)
account.withdraw(300)
account.check_balance()
```

Output

```
Deposited: 500
Withdrawn: 300
Current Balance: 1200

=== Code Execution Successful ===
```

Overall Conclusion

This lab demonstrates how AI-assisted code completion helps in generating structured, readable, and correct Python programs. Human review is essential to ensure correctness and efficiency.